

MUHAMMAD HAMMAD

Mansehra - KPK

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Education

Hazara University

Bachelor of Software Engineering

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Mansehra, KPK

About Me

- Software Engineering student specializing in backend API development with Python, FastAPI, and Flask. Experienced in building and designing RESTful APIs with PostgreSQL, Redis caching, async task queues, and JWT authentication. Familiar with system design fundamentals and production-grade backend architecture.

Projects

AI-Powered URL Shortener API | *Python, FastAPI, PostgreSQL, Redis, SQLAlchemy, ARQ* | [GitHub](#)

- Designed a two-layer Redis caching strategy (in-memory L1 + persistent L2) to serve repeated redirect lookups without hitting PostgreSQL, reducing database read load on hot URLs and keeping p99 latency under 10ms.
- Offloaded AI slug generation (Groq API) to an async ARQ background task queue, decoupling slow LLM inference from the request cycle and returning API responses in under 50ms regardless of model latency.
- Implemented JWT-based authentication with per-user rate limiting enforced at the middleware layer, preventing abuse while isolating quota tracking per identity.
- Built URL expiry with automatic cache invalidation: on expiry, both Redis layers are flushed atomically so stale redirects are never served — eliminating the cache-DB consistency problem.
- Scheduled periodic cleanup of expired URL records using APScheduler as a background job, keeping the PostgreSQL table lean without external cron infrastructure.

LSC – Two-Sided Marketplace Platform | *Python, Flask, MySQL, HTML5, CSS, JavaScript, Jinja2* | [GitHub](#)

- Built a full-stack two-sided marketplace connecting service seekers with local providers across 6 professions including plumbing, electrical, and cleaning services.
- Implemented role-based access control with three distinct user types (seekers, providers, admins) each with separate dashboards, permissions, and booking workflows.
- Designed real-time messaging system between customers and service providers with booking status tracking and history.
- Architected authentication system using Bcrypt password hashing and parameterized SQL queries preventing SQL injection attacks.
- Built booking engine supporting both direct provider selection and general service requests with status management.

Perfumania – E-Commerce Store | *Python, FastAPI, PostgreSQL, SQLAlchemy, Jinja2* | [GitHub](#) | [Live](#)

- Built and deployed a production e-commerce application for a real client, live at perfumania-shop.up.railway.app, with product browsing, cart management, and order tracking.
- Implemented secure authentication using Bcrypt and Argon2 password hashing with session management, protecting customer accounts and admin operations.
- Designed product and order management system with PostgreSQL and SQLAlchemy ORM supporting collections, inventory, and per-order status tracking.
- Integrated transactional email notifications via fastapi-mail for order confirmations and account registration, automating customer communication.
- Deployed on Railway with environment-based configuration using pydantic-settings, cleanly separating development and production secrets.

Technical Skills

Languages: Python, SQL, HTML/CSS

Frameworks: FastAPI, Flask

Databases: PostgreSQL, MySQL

Tools: Git, GitHub, VS Code, Postman

Technologies: Redis, Docker